

Himanshu Ambalal Patil

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SUMMARY

Backend Software Engineer with 3+ years of experience designing scalable, data-driven systems used by global enterprise customers. Improved query and API performance by up to 10× and automated workflows to reduce manual effort by 50–90% using Java, Spring Boot, Python, SQL, MongoDB, and CI/CD pipelines. Seeking Software Engineering roles focused on performance, scalability, and clean system design.

EDUCATION

University at Buffalo, SUNY | Master of Science: Computer Science and Engineering **August 2024 - December 2025**

Course work: Data Intensive Computing, Introduction to Machine Learning, Deep Learning, Intro to Pattern Recognition, DMQL

SGGSIE&T, India | Bachelor of Technology: Computer Science and Engineering **August 2017 - July 2021**

Course work: OOP with Java, Data Structures, ADBMS, Software Engineering, Big Data Analytics, Probability and Statistics.

TECHNICAL SKILLS

- **Programming & Development:** Java, Python, C, SQL, JavaScript, HTML, CSS, Spring Boot, Spring MVC, Hibernate.
- **Databases:** MongoDB, PostgreSQL, MySQL (SQL & NoSQL design, query optimization)
- **Data & Analytics:** Machine Learning, Deep Learning, NLP, Pattern Recognition, Predictive Modelling, Data Structures & Algorithms, KPI Analysis.
- **DevOps & Tools:** CI/CD, Jenkins, Docker, Git, GitHub, Jira, Jupyter, VS Code, MongoDB Compass, Eclipse, IntelliJ, Redash, Power BI.

WORK EXPERIENCE

Kaiburr | Software Engineer **April 2022 – July 2024**

- Developed Java backend microservices using Spring Boot, JPA, and MongoDB to ingest metadata from tools like Jira, ADO, Bitbucket, and Jenkins, processing data for over 100+ projects across enterprise clients while applying SOLID principles for clean and maintainable code.
- Engineered backend analytics workflows using Groovy and Python, translating customer-defined business logic and rules into configurable, reusable components that powered enterprise dashboards used by product, engineering, and operations teams.
- Led performance optimization for parent–child analytics dashboards by refactoring MongoDB aggregation pipelines and data models, standardizing a single extensible query framework that powered multi-filter, reusable dashboards across products and delivered a 99% reduction in query execution time.
- Authored and standardized complex MongoDB aggregation pipelines, powering multi-filter, reusable dashboards through a single extensible query framework used across products.
- Reduced dashboard development effort by 90%, introducing parameterized queries that dynamically adapted to new filters, KPIs, and use cases without additional backend changes.
- Improved query and dashboard performance by up to 10×, implementing targeted indexing strategies, aggregation refactoring, and query reuse across high-traffic dashboards.
- Automated query generation using Python-based templating, cutting manual configuration effort by 50% and reducing production misconfigurations and runtime errors.
- Identified and resolved backend and data-layer bottlenecks affecting latency and scalability, ensuring low-latency analytics and reliable dashboard rendering under increasing data volume.
- Recognized within 6 months as the primary owner for analytics performance and MongoDB optimization, mentoring new joiners, assigning and reviewing tasks, establishing team-wide best practices, and delivering technical training on scalable query design and dashboard architecture.

Cognizant Technology Solutions | Programmer Analyst Trainee **August 2021 - March 2022**

- Implemented and maintained RESTful APIs and backend services using Java, Spring Boot, Spring MVC, and Hibernate for enterprise-grade applications.
- Developed full-stack web features using HTML, CSS, and JavaScript, enhancing UI consistency and improving user experience.
- Modeled and tuned MySQL schemas, indexes, and queries to enable scalable and reliable backend data access.
- Applied advanced Core Java concepts including OOP, Collections, Multithreading, and Exception Handling to solve 2+ real-world case study problems.
- Collaborated within Agile/Scrum teams to debugging, unit testing, and backend performance tuning, supporting stable sprint deliveries.

PROJECT

Snap2Vibe - AI-Powered Photo-to-Music Recommender([Link](#)) **August 2025 - December 2025**

- Built an end-to-end photo-to-music recommendation system using OpenCLIP (ViT-B/32), FastAPI/Flask, React, and Spotify Web API for real-time interaction.
- Delivered personalized song recommendations through vision-based mood and scene analysis with low-latency backend integration.

KnowYourDoc | Hackathon Project **February 2025 - February 2025**

- Architected an end-to-end legal document analysis platform using FastAPI, PyPDF2, Gemini API, and a responsive HTML/CSS/JavaScript frontend to support PDF uploads, text extraction, and interactive insight visualization.
- Reduced manual document review effort by ~60% by implementing robust JSON parsing and validation logic to surface structured outputs such as summaries, complexity ratings, red flags, extracted figures, and loopholes.

F1 Race Winner Predictor | Machine Learning ([Link](#)) **August 2024 - December 2024**

- Constructed a machine learning pipeline to predict Formula 1 race winners using historical Grand Prix data, including driver performance, team metrics, grid positions, and pit stop strategies.
- Trained and evaluated five models (Logistic Regression, KNN, Random Forest, XGBoost, Neural Network), achieving best recall of 0.78 with XGBoost through GridSearchCV tuning and cross-validation to address class imbalance.